

Model Analysis Report

Building Seismic-Resilience Credit Model

MKM-SEIS-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Seismic credit by source-to-site distance R_{JB} (RC pre-code, Moderate zone, all BRI absent)

R_{JB}	Median PGA (g)	No-collapse (%)	Loss freq (%)	Spread (bps)
2 km	0.1157	93.1	0.1228	5.00
5 km	0.0835	94.4	0.0954	4.00
10 km	0.0520	97.9	0.0526	1.50
20 km	0.0291	100.0	0.0151	0.0000
50 km	0.0129	100.0	0.0088	0.0000
100 km	0.0068	100.0	0.0083	0.0000

Table 2: Seismic spread (bps) by $R_{JB} \times$ construction type (Moderate zone, all BRI absent)

R_{JB}	RC pre-code	Masonry URM	RC modern	Steel MRF
2 km	5.00	9.50	3.50	2.50
10 km	1.50	3.00	0.0000	0.0000
20 km	0.0000	0.5000	0.0000	0.0000
50 km	0.0000	0.0000	0.0000	0.0000

Table 3: Seismic credit by hazard zone (RC pre-code, $R_{JB} = 20$ km, all BRI absent)

Zone	No-collapse (%)	Loss freq (%)	Spread (bps)
Very low	100.0	0.0012	0.0000
Low	100.0	0.0061	0.0000
Moderate	100.0	0.0151	0.0000
High	99.1	0.1067	2.50
Very high	99.4	0.2868	4.50

Table 4: Seismic spread (bps) by hazard zone $\times$ soil type (RC pre-code, $R_{JB} = 20$ km) — the soft-soil amplification and GS02 gate

Zone	Rock (A)	Stiff (B)	Soft (D) no GS02	Soft (D) GS02
Low	0.0000	0.0000	0.0000	0.5000
Moderate	0.0000	0.0000	2.50	2.50
High	2.00	2.50	9.50	4.00
Very high	5.00	4.50	33.0	21.5

Table 5: Seismic credit by BRI resilience level (RC pre-code, Moderate zone, near-fault $R_{JB} = 5$ km)

BRI level	No-collapse (%)	Loss freq (%)	Spread (bps)
NR	94.4	0.0954	4.00
B	97.2	0.0645	2.00
A	99.3	0.0445	0.5000
AA	100.0	0.0226	0.0000

Table 6: Seismic credit by individual BRI measure (single-measure sweeps; Moderate zone, near-fault $R_{JB} = 5$ km, RC pre-code, stiff soil)

Measure	No-collapse (%)	Spread (bps)	Δ spread
None (baseline)	94.4	4.00	0.0000
GS08: Lateral force resistance	95.8	3.00	-1.00
GS09: Connected / braced	95.8	3.00	-1.00
GS12: Base isolation	97.2	2.00	-2.00
GS14: No vertical irregularities	95.1	3.50	-0.5000
GS02: Seismic foundation (F_a gate)	94.4	4.00	0.0000

Table 7: Seismic spread (bps) by hazard zone $\times$ BRI resilience level (RC pre-code, $R_{JB} = 20$ km) — strong measures are most valuable in high-hazard zones

Zone	NR	B	A	AA
Very low	0.0000	0.0000	0.0000	0.0000
Low	0.0000	0.0000	0.0000	0.0000
Moderate	0.0000	0.0000	0.0000	0.0000
High	2.50	2.00	2.00	0.0000
Very high	4.50	6.00	3.50	1.00

Table 8: Seismic spread (bps) by $R_{JB} \times$ BRI resilience level (RC pre-code, Moderate zone)

R_{JB}	NR	B	A	AA
2 km	5.00	4.00	2.00	0.5000
10 km	1.50	0.5000	0.0000	0.0000
20 km	0.0000	0.0000	0.0000	0.0000
50 km	0.0000	0.0000	0.0000	0.0000

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 9: Stress testing scenarios for Building Seismic-Resilience Credit Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 10: Performance metrics for Building Seismic-Resilience Credit Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.